
Flipkart Review Sentiment Analysis

Leveraging Machine Learning for Customer
Insights



PROJECT OVERVIEW

Objective

Build a reliable ML model to classify Flipkart product reviews into positive or negative sentiments to speed up feedback analysis.

Business Value

Automated sentiment analysis helps teams quickly understand customer satisfaction and pinpoint areas for improvement.

Technology Stack

Uses Python, Pandas, Scikit-Learn and NLTK, with Matplotlib/Seaborn for visuals.



DATASET PROFILE

DATA SOURCE

The analysis is grounded in a dataset comprising **2,304 individual reviews** sourced from Flipkart, covering a diverse range of electronics including laptops and monitors.

Diverse linguistic styles and product-specific terminology.

KEY FEATURES

Product Name: The item being reviewed.

Review: Textual feedback provided by the customer.

Rating: Numerical score from 1 to 5 stars.

INITIAL STATE

Initial inspection revealed a need for systematic cleaning to address **missing values** and standardize data formats, ensuring the integrity of the modeling phases.

Crucial foundation for subsequent analysis.

DATA REFINEMENT

Handling Missing Data

To maintain high data integrity, a rigorous cleaning process was implemented where any rows containing null values in the critical 'Review' or 'Rating' columns were removed.

Type Standardization

Ratings were standardized as integers to facilitate both numerical analysis and the creation of sentiment labels, ensuring consistency across the dataset.

**OUTCOME: A CLEAN,
STRUCTURED DATASET
PROVIDING A RELIABLE
FOUNDATION FOR
EXPLORATORY ANALYSIS AND
MODELING.**



EXPLORATORY DATA ANALYSIS: RATING PATTERNS

Rating Distribution

Visual analysis revealed the frequency of each rating from 1 to 5 stars, providing a clear overview of customer satisfaction across the dataset.

Customer Behavior

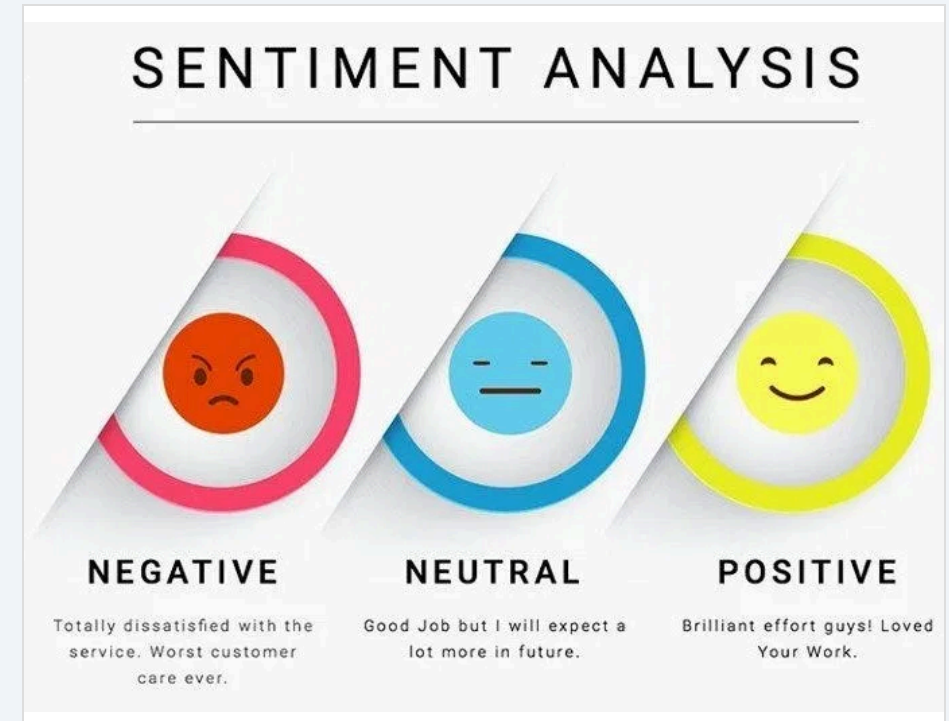
Feedback clusters at the extremes (1 or 5 stars), where customers are most motivated to review after exceptional or poor experiences.

Data Balance

Understanding the distribution is vital for identifying class imbalances, guiding the definition of positive and negative sentiment labels.

Strategic Insight

The distribution provides a baseline for expectations, helping define the threshold for successful product performance.



SENTIMENT LABELING: DEFINING SUCCESS AND FAILURE

Classification Logic

To create a clear binary classification task, ratings were mapped to specific sentiment labels:

Positive (1): Ratings of 4 and 5 stars.

Negative (0): Ratings of 1 and 2 stars.

Neutral Exclusion

Reviews with a **3-star rating** were excluded from the analysis. This strategic decision focuses the model on definitive sentiments, removing ambiguous feedback that could degrade classification accuracy.

Target Variable Definition

The resulting '**Sentiment**' column serves as the ground truth for the machine learning model. By simplifying the five-star scale into a binary format, we establish a clear objective for the algorithm: distinguishing between customer satisfaction and dissatisfaction.

The final distribution showed a higher proportion of positive reviews, reflecting a generally high level of customer satisfaction among the sampled products on the Flipkart platform.

TEXT PREPROCESSING

01

Normalization

All text was converted to lowercase to ensure uniformity across the entire dataset.

02

Noise Removal

Special characters, numbers, and punctuation were stripped to focus on semantic content.

03

Stopword Filtering

Common English words (e.g., "the", "is") were removed using NLTK's standard list.

04

Stemming

Applied SnowballStemmer to reduce words to their root form, reducing feature dimensionality.

Raw text reviews underwent extensive preprocessing to transform them into a structured format suitable for machine learning, ensuring that each review contributes meaningfully to the classification task.

OUTLIER MANAGEMENT

Length Analysis

We checked word counts to flag unusually short or long reviews that could harm model learning.

Short Review Removal

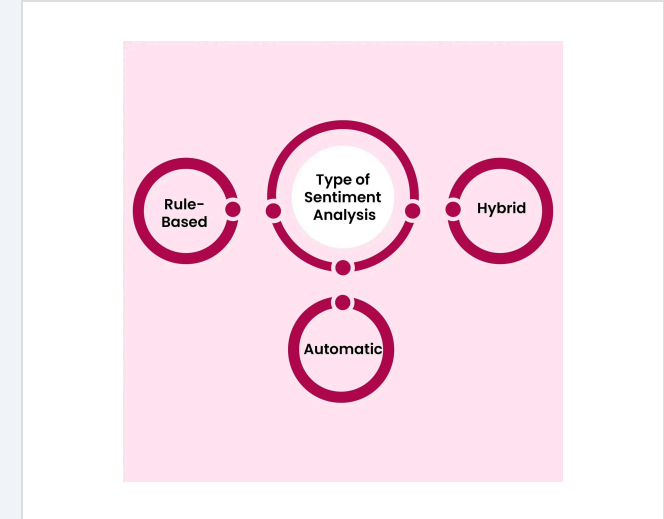
Reviews under three words were removed because they lack meaningful sentiment signals.

Long Review Capping

Very long reviews (above the 95th percentile) were filtered to avoid skewing the model and to save compute.

Result

The dataset became more consistent, improving reliability of the sentiment classifier.



FEATURE ENGINEERING: TF-IDF VECTORIZATION

Methodology

Used Term Frequency-Inverse Document Frequency (TF-IDF) to convert cleaned text into numerical vectors, capturing the semantic importance of words.

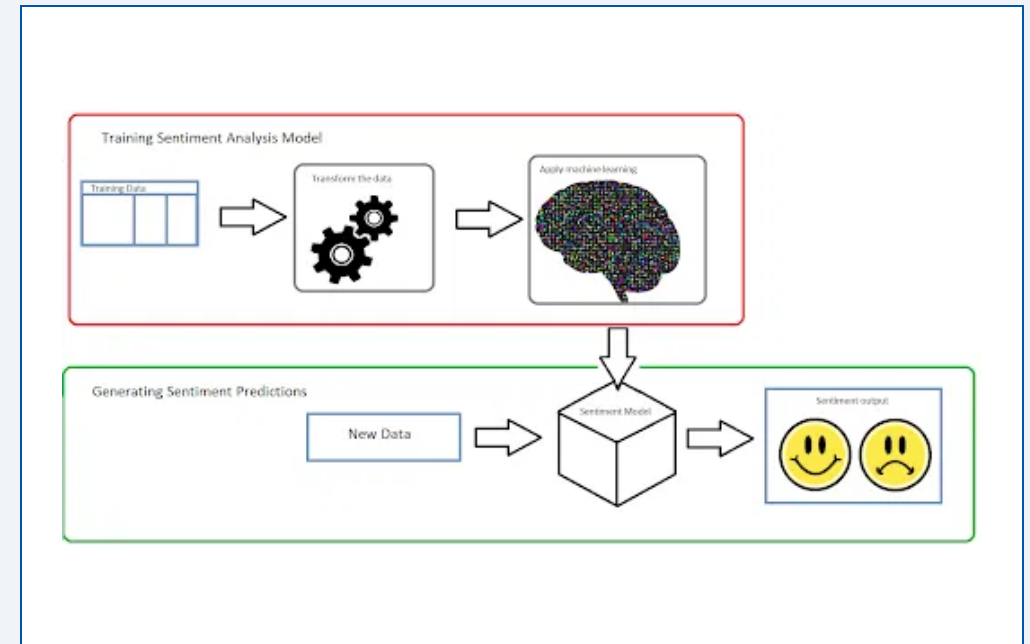
Feature Selection

Limited the vocabulary to the top 5,000 most significant features to optimize model performance and prevent overfitting on rare terms.

Data Split

The processed features were divided into training (80%) and testing (20%) sets for unbiased evaluation of the model's performance.

OBJECTIVE: TRANSFORM QUALITATIVE FEEDBACK INTO QUANTITATIVE DATA FOR MACHINE LEARNING ANALYSIS.



MODEL DEVELOPMENT

Algorithm Choice

Logistic Regression was chosen for its efficiency and interpretability, delivering strong results on binary text classification tasks for sentiment detection.

Training Process

The model was trained on TF-IDF vectors using the 'liblinear' solver, effective for medium-sized datasets with high-dimensional features.

Reproducibility

A fixed random state of 42 ensured consistent results across runs, enabling reliable model comparison and validation.

Model Fit

The algorithm learned relationships between word patterns and customer sentiments, producing a dependable predictive model.



PERFORMANCE EVALUATION: 91.92% ACCURACY

91.92%

OVERALL ACCURACY

The model demonstrated exceptional performance on the unseen test dataset.

1.00

POSITIVE RECALL

The model successfully identified all positive reviews in the test set.

1.00

NEGATIVE PRECISION

Every review flagged as negative was indeed negative, ensuring high reliability.

CLASSIFICATION REPORT

SENTIMENT	PRECISION	RECALL	F1-SCORE
Negative	1.00	0.29	0.45
Positive	0.92	1.00	0.96
Accuracy	-	-	0.92
Weighted Avg	0.93	0.92	0.90

The model is exceptionally strong at identifying positive sentiment. The lower recall for negative reviews suggests a need for more negative training samples to improve detection of subtle dissatisfaction.

CONCLUSION & FUTURE WORK

Project Summary

Built a high-accuracy sentiment classifier for Flipkart reviews with 91.92% accuracy, enabling scalable monitoring of customer satisfaction.

Key Takeaways

Reliability

Strong detection of positive sentiment and accurate negative flagging.

Efficiency

Logistic Regression with TF-IDF is a robust, interpretable baseline for text classification.

FUTURE ENHANCEMENTS

- **Contextual Features:** Add N-grams to capture richer context.
- **Advanced Architectures:** Evaluate LSTM or Transformer models (e.g., BERT).
- **Real-time Deployment:** Expose the model via a live API for automated feedback monitoring.
- **Data Augmentation:** Expand negative samples to address class imbalance.